

Yield, Duration, Convexity, and the Yield Curve

LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- **Explain price sensitivity.** Use a fixed-income cash-flow model to establish the direction and curvature of its response to a yield change.
- **Compute local risk measures.** Calculate modified duration and convexity and use them to approximate a price change.
- **Read the yield curve.** Distinguish yield to maturity, par yield, and spot rates and explain which rates price individual cash flows.

1 A Safe Payment Can Have a Risky Price

L2a treated promised Treasury payments as known, but a holder who sells before maturity receives a market price. That price changes whenever the return available on comparable new securities changes. A fixed promise and a fluctuating resale value are therefore entirely compatible.

The 2023 failure of Silicon Valley Bank shows why bond-price risk matters. The bank held long-duration fixed-rate assets and relied heavily on uninsured deposits. Rising rates reduced the fair value of its securities. When depositors withdrew funds, the bank had to confront losses that its hold-to-maturity accounting had kept out of view. Falling bond values and deposit withdrawals reinforced each other, so interest-rate risk was important but was not the only cause of the failure.

Burton Malkiel's bond-pricing principles summarize how bond prices change: price moves opposite yield, the price-yield curve is convex, longer maturity generally increases percentage sensitivity, and a larger coupon generally reduces it. These statements assume an option-free bond with fixed cash flows and hold the other contract terms fixed.

2 Which Variables Move the Price?

Before restricting attention to the yield, it is worth asking which variables move a bond price at all. A zero-coupon Treasury bill answers that question cleanly, because a single promised payment gives a price that is smooth in both arguments. Let $V_P > 0$ dollars denote the face value paid at maturity, let $T > 0$ years denote the remaining time to maturity, let the positive integer n denote the number of compounding intervals per year, and let y denote the nominal annual yield with $1 + y/n > 0$. Holding n fixed, the bill price is $V_B(y, T) = V_P (1 + y/n)^{-nT}$, and its total differential is given by Eq. 1:

$$dV_B = \underbrace{\left(-V_P (1 + y/n)^{-nT} \frac{T}{1 + y/n} \right)}_{\partial V_B / \partial y} dy + \underbrace{\left(-V_P (1 + y/n)^{-nT} n \log(1 + y/n) \right)}_{\partial V_B / \partial T} dT. \quad (1)$$

The calculation follows the same procedure used for thermodynamic state functions: list the independent variables and differentiate with respect to each one. Here y is a market variable, while T is a contract term, so changes in y and T represent different comparisons.

The coupon rate is a third possible variable. In the coupon-bearing family with fixed payment count $N = nT$, $\partial V_B / \partial c = (V_P / n) \sum_{j=1}^N \mathcal{D}_{j,0}^{-1}(y; n) > 0$, even at $c = 0$. A bill has no coupon term in Eq. 1 because its contract fixes $c = 0$ and therefore $dc = 0$. The coupon derivative itself is not zero.

The maturity derivative needs two qualifications. First, $\partial V_B / \partial T < 0$ requires $y > 0$. At $y = 0$, $\log(1 + y/n) = 0$, so maturity does not change the price. For $-n < y < 0$, the sign reverses.

Second, dT can describe two different comparisons. Changing the contractual maturity compares two securities. The passage of time changes the remaining maturity of one security, giving $dT = -dt$. Both changes are smooth for a bill because its price depends on T only through the exponent. A coupon security is different: its dirty price changes smoothly between coupon dates, but the payment schedule changes when a coupon leaves the schedule.

A total differential is linear, so Eq. 1 describes first-order changes but not curvature. For the rest of these notes, we hold the cash-flow schedule fixed, vary only the yield, and include the second derivative.

3 Price as a Function of Yield

To isolate interest-rate risk, hold the promised cash flows fixed and vary only the yield used to discount them.

Defn. 3.1 defines the resulting bond price.

Definition 3.1. Yield-based bond price

Let the positive integer n denote the number of compounding intervals per year, let the positive integer N denote the number of remaining compounding periods, and let $X_j \geq 0$ dollars be the fixed cash flow paid at period index $j \in \{1, \dots, N\}$, with at least one $X_j > 0$. Let the nominal annual yield to maturity be y with $1 + y/n > 0$. Under a flat-yield convention, the time-0 dirty price $V_B(y)$, meaning the value including any accrued interest, is given by Eq. 2:

$$V_B(y) := \sum_{j=1}^N X_j \left(1 + \frac{y}{n}\right)^{-j}. \quad (2)$$

The yield y is the single internal discount rate applied to every promised payment; it is not a complete term structure of spot rates. Permitting $X_j = 0$ on periods carrying no payment lets a zero-coupon bill share this definition, with $X_N = V_P$ its only nonzero flow. The integer exponents place the valuation date on a coupon date; a general settlement date requires fractional-period timing conventions that these notes do not need.

The j -period discount factor $\mathcal{D}_{j,0}^{-1}(y; n) = (1 + y/n)^{-j}$ converts the notation in Eq. 2 into a sum of nonnegative discounted claims, at least one of which is strictly positive. Differentiating that sum establishes both the direction and curvature of the price response (Thm. 3.1).

Theorem 3.1. Monotonicity and convexity of an option-free bond

Under the assumptions of Defn. 3.1, assume $1 + y/n > 0$. The first and second derivatives of the dirty price with respect to the nominal annual yield are given by Eq. 3:

$$\frac{dV_B}{dy} = -\frac{1}{n} \sum_{j=1}^N jX_j (1 + y/n)^{-j-1} < 0, \quad \frac{d^2V_B}{dy^2} = \frac{1}{n^2} \sum_{j=1}^N j(j+1)X_j (1 + y/n)^{-j-2} > 0. \quad (3)$$

The strict inequalities hold because no term of either sum is negative and at least one term of each is positive. Consequently, the bond price is strictly decreasing and strictly convex as a function of yield. The result assumes a fixed, nonnegative, and not identically zero cash-flow schedule, and therefore does not automatically apply to securities whose cash flows change when rates change, such as callable bonds.

The signs in Eq. 3 explain the shape of the price-yield curve. A higher yield lowers every discounted payment. Positive curvature makes the gain from a yield decrease larger than the loss from an equal yield increase, measured from the same starting yield (Fig. 1).

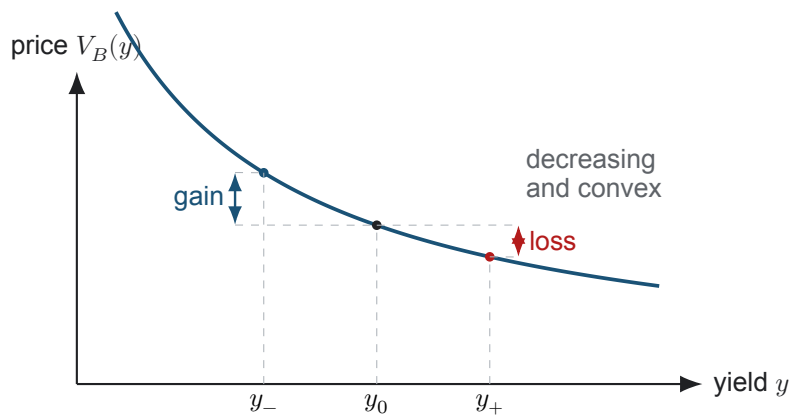


Figure 1: The price-yield curve for a fixed positive cash-flow stream. The comparison yields are $y_- := y_0 - \Delta y$ and $y_+ := y_0 + \Delta y$ for an equal change $\Delta y > 0$ from the initial yield y_0 . Convexity makes the vertical price gain from the rate decrease exceed the price loss from the equal rate increase.

The bill-sensitivity notebook tests the first part of Thm. 3.1 using zero-coupon Treasury auction data. With only one maturity payment, the relationship is easy to see.

COMPANION NOTEBOOK**L2b: Treasury Bill Price Sensitivity →**

Measure how a zero-coupon bill price changes with yield and remaining time using Treasury auction records.

4 Duration and Convexity

The derivative signs answer *which direction*; risk management also needs *how much*. Normalizing the derivatives by price produces two local measures with interpretable units (Defn. 4.1).

Definition 4.1. Modified duration and convexity

For the fixed cash-flow schedule, nominal annual yield y , coupon frequency n , and price $V_B(y) > 0$ defined in Defn. 3.1, the modified duration $D_{\text{mod}}(y)$ and convexity $K(y)$ are defined by Eq. 4:

$$D_{\text{mod}}(y) := -\frac{1}{V_B(y)} \frac{dV_B}{dy}, \quad K(y) := \frac{1}{V_B(y)} \frac{d^2V_B}{dy^2}. \quad (4)$$

When the yield y is expressed as a decimal per year, modified duration has units of years and convexity has units of years squared. Both measures are positive under the assumptions of Thm. 3.1.

For a small yield shock Δy , a second-order Taylor approximation combines the two normalized sensitivities. The approximate fractional price change is given by Eq. 5:

$$\frac{\Delta V_B}{V_B} \approx -D_{\text{mod}} \Delta y + \frac{1}{2} K (\Delta y)^2. \quad (5)$$

The duration term supplies the local slope, while the convexity term corrects the straight-line approximation for curvature. This formula uses a decimal yield change: a 25-basis-point shock is $\Delta y = 0.0025$, where one basis point is 10^{-4} .

5 How Contract Structure Changes Sensitivity

The *Macauley duration* D_{mac} is the present-value-weighted average time to payment, and the modified duration of Defn. 4.1 is that average discounted by one period, $D_{\text{mod}} = D_{\text{mac}} / (1 + y/n)$. Because the two differ only by that positive factor, both respond to payment timing in the same way: moving more value toward distant dates increases exposure to discount-rate changes, and moving value toward earlier dates reduces it. That principle gives the carefully qualified maturity and coupon comparisons associated with Malkiel.

Maturity effect. Assume $y > 0$ throughout this comparison. Malkiel's maturity statements compare bonds selling at par, and that restriction supplies a closed form. Summing the geometric series in Eq. 2 for a coupon rate c gives $V_B/V_P = c/y + (1 - c/y)(1 + y/n)^{-N}$. Differentiating with respect to y at fixed c , and only then setting $c = y$ so that $V_B = V_P$, yields the par-bond modified duration and its increment, given by Eq. 6:

$$D_{\text{mod}}(N) = \frac{1 - (1 + y/n)^{-N}}{y}, \quad D_{\text{mod}}(N + 1) - D_{\text{mod}}(N) = \frac{(1 + y/n)^{-N-1}}{n}. \quad (6)$$

The first expression increases with N but remains below $1/y$, the modified duration of a perpetuity. The second expression shows that each additional period adds less duration than the previous one. These are Malkiel's maturity results.

For par bonds, the initial price is V_P at every maturity. The dollar sensitivity $-dV_B/dy = V_P D_{\text{mod}}(N)$ therefore has the same ranking as the percentage sensitivity. This agreement need not hold away from par.

For example, the modified duration of a zero-coupon security grows without bound as T increases. Its dollar sensitivity, $-dV_B/dy = V_P T (1 + y/n)^{-nT-1}$, first rises and then falls. It reaches its maximum at $T = 1 / (n \log(1 + y/n))$ because the declining price eventually outweighs the growing duration. These statements concern sensitivity to yield; they do not say that adding another coupon period always lowers the bond's price.

Coupon effect. Holding yield, maturity, face value, and coupon frequency fixed, a higher coupon shifts a larger fraction of present value toward earlier dates. The resulting percentage price sensitivity is therefore lower. The

comparison requires at least two remaining payment dates: if a single payment remains, all value arrives on one date and the modified duration is $1/(n(1 + y/n))$ years regardless of the coupon.

Discrete-maturity caution. A coupon count N is an integer. Extending maturity adds a payment and changes the cash-flow vector, so differentiating a finite coupon sum with respect to a continuous maturity variable can be misleading. Compare two explicitly defined cash-flow streams or use duration with the cash flows held fixed.

6 From One Yield to a Curve of Discount Rates

Yield to maturity summarizes a security's price and promised cash flows with one internal rate of return. That rate is useful for quoting and comparison, but it is not a complete set of discount rates. A *spot rate* applies to a single payment at one maturity; the resulting spot curve supplies one discount factor for each payment date. A *par yield* is the coupon rate that makes a hypothetical coupon security trade at par under that discount curve.

For cash flows X_j paid at times t_j , a term-structure valuation replaces the flat-yield expression in [Eq. 2](#) with

$$V_B = \sum_{j=1}^N X_j D(0, t_j), \quad (7)$$

where $D(0, t_j)$ is the time-0 discount factor for maturity t_j . The curve's level, slope, and curvature summarize how required compensation varies across maturities; they do not by themselves identify a unique forecast of future short rates. This distinction prepares us for the multi-period asset models used in Week 3.

The coupon-security notebook uses Treasury note data to show how price changes with yield and coupon timing.

COMPANION NOTEBOOK**L2b: Treasury Duration and Convexity** →

Price a coupon Treasury security and compare its yield, coupon, duration, and convexity sensitivities.

Optional advanced paths

Students who want the full theorem proofs, empirical curve exercises, stochastic short-rate models, or STRIPS construction can continue in [week-2/L2b/advanced/](#). These extensions are deliberately separated from the required sequence so that Week 3 can begin with equity markets.

KEY TAKEAWAYS

In this lecture, we differentiated a fixed cash-flow bond price, established its decreasing convex price-yield shape, and converted those derivatives into duration and convexity risk measures.

- **Yield and price oppose.** For a fixed positive cash-flow stream, a higher yield lowers price because every promised payment is discounted more heavily.
- **Curvature creates asymmetry.** Positive convexity makes the gain from a yield decrease larger than the loss from an equal yield increase around the same starting yield.
- **A yield is not a curve.** Payment timing controls duration; yield to maturity is a single internal rate, while spot discount factors price individual cash flows and par yields describe hypothetical securities priced at par.

Next, carry these return, risk, and time-scale ideas into equity markets and empirical growth-rate data.

ADDITIONAL READING

- Burton G. Malkiel, “Expectations, Bond Prices, and the Term Structure of Interest Rates,” *Quarterly Journal of Economics* 76(2), 197–218 (1962).
- Federal Reserve Board, *Review of the Federal Reserve’s Supervision and Regulation of Silicon Valley Bank* (2023), for the interaction of interest-rate, liquidity, and governance risk.
- Frank J. Fabozzi and Francesco A. Fabozzi, *Bond Markets, Analysis, and Strategies*, 10th ed. (2021), for duration and convexity conventions.

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